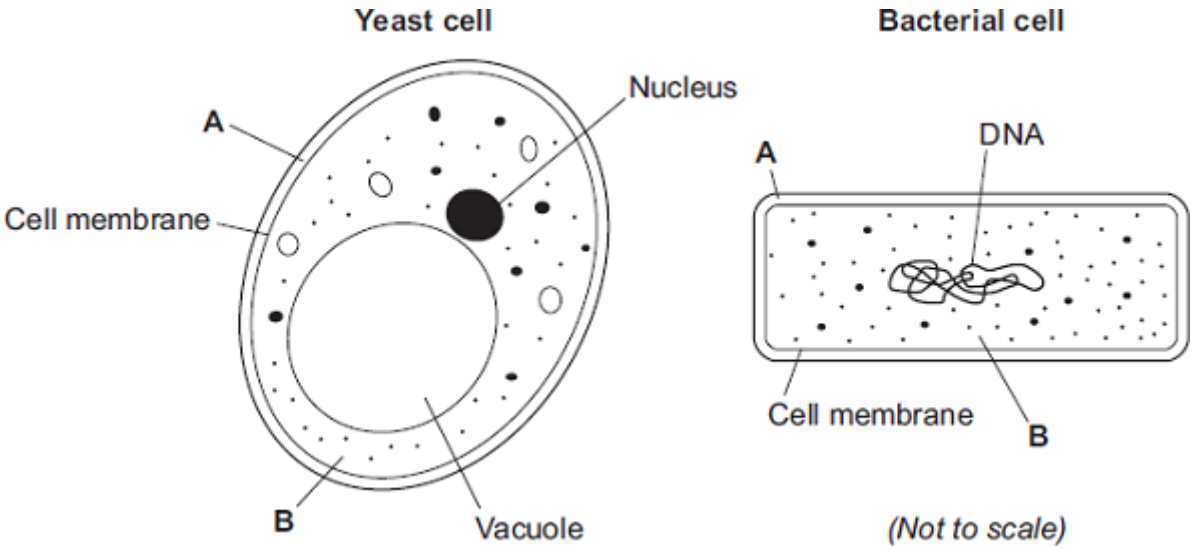
Q1. (a) The diagrams show the structures of a yeast cell and a bacterial cell.



(i) Both the yeast cell and the bacterial cell have structures **A** and **B**.

Name structures **A** and **B**.

A .....

B .....

(2)

(ii) The yeast cell and the bacterial cell have different shapes and sizes.

Give **one** other way in which the structure of the bacterial cell is different from the structure of the yeast cell.

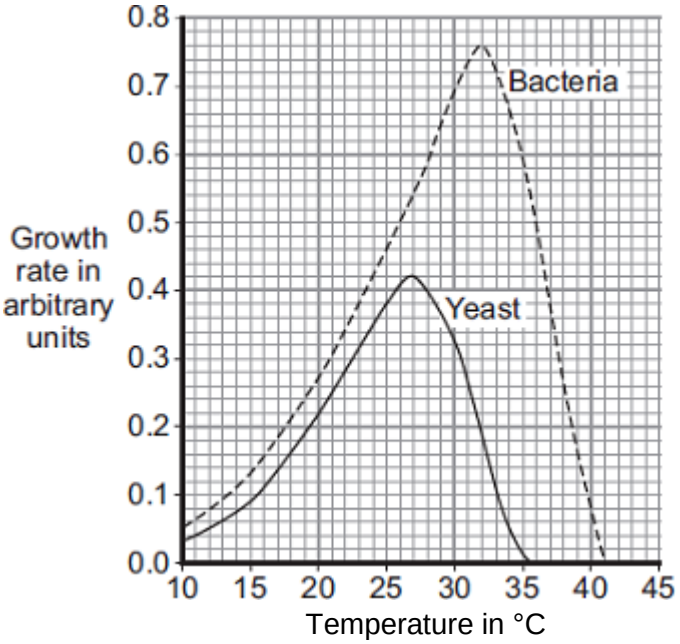
.....

.....

(1)

(b) Sourdough bread is light in texture and tastes slightly sour. The bread is made using two types of microorganism, a yeast and a bacterium. The bacterium can make acids such as lactic acid. The acid makes the bread taste sour.

The graph shows how the growth rates of the yeast and the bacteria change with temperature.



(i) Sourdough bread rises fastest at 27°C.
Use information from the graph to explain why.

.....

.....

.....

.....

(2)

(ii) The bread tastes most sour if it rises at 32°C.
Use information from the graph to explain why.

.....

.....

.....

.....

(2)

(Total 7 marks)

Mark schemes

Q1.

(a) (i) A = (cell) wall

ignore cellulose

1

B = cytoplasm

1

(ii) any **one** from:

accept has DNA instead of a nucleus, but not just has DNA

- bacterial cell / it has no nucleu>

  allow no mitochondria

- DNA free in cytoplas>

  ignore size

- has no vacuole / no vesicle>

  ignore strands of DNA

1

(b) (i) <u>yeast</u> grows best / better / well **or** optimum temperature for <u>yeast</u> / more <u>yeast</u> present

allow <u>yeast</u> works best / better / well

1

(yeast) makes CO_2 **or** respires / respiration

allow fermentation

1

(ii) <u>bacterium</u> grows best / better / well / more <u>bacteria</u> present **or** optimum temperature for <u>bacterium</u>

ignore microorganisms / microbes

allow works / respires best / better / well

1

(bacterium) makes (lactic) acid

*do **not** allow wrong acid*

1

[7]